



1.4 Financial mathematics: loans and amortization

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Compound growth follows $FV = PV(1+i)^n$ where i is the interest rate per compounding period. A loan is repaid through amortization: each fixed regular payment (PMT) covers the interest owed on the outstanding balance, with the remainder reducing the principal, so the interest portion of each payment shrinks over time. Financial (TVM) solvers relate N (number of payments), $I\%$ (annual rate), PV , PMT and FV , with $PMT = PV \times i / (1 - (1+i)^{-n})$ for a fully amortized loan ($FV=0$).

Worked example: follow the reasoning

A loan of \$12,000 is taken at a nominal 6% p.a. compounded monthly, to be repaid in equal monthly instalments over 3 years (36 months). Find the monthly payment.

Solution: $i = 0.06/12 = 0.005$, $n=36$. $PMT = 12000 \times 0.005 / (1 - 1.005^{-36}) = 60 / (1 - 0.8357) = 60 / 0.1643 \approx \365.10 .

Quick check · multiple choice

In a TVM/finance solver used for a loan, what does PMT represent?

- A The total interest paid
- B The regular payment amount per compounding period
- C The present value of the loan
- D The number of payments

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Camila takes out a loan of \$8,500 at a nominal annual interest rate of 9%, compounded monthly, to be repaid with equal monthly payments over 2 years. [6]
(a) Find the monthly payment, to the nearest cent. (b) Find the total interest paid over the life of the loan, to the nearest cent.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $i=0.09/12=0.0075$, $n=24$. $PMT = 8500 \times 0.0075 / (1 - 1.0075^{-24}) = 63.75 / (1 - 0.8358) = 63.75 / 0.1642 \approx \388.44 . (b) Total repaid = $388.44 \times 24 \approx \9322.56 ; interest = $9322.56 - 8500 \approx \$822.56$.

Quick check answer

B. PMT is the fixed regular payment made each compounding period.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics: AI subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.